Bihar Engineering University, Patna

B.Tech. 7th Semester Examination, 2023

Course: B.Tech.
Code: 110728

Subject: Electrical Safety

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): [2 x 7 = 14]

(a) What is the purpose of grounding electrical systems?
(i) To provide a path for current to flow
(ii) To protect against electric shocks.
(iii) To reduce electrical resistance
(iv) To increase energy efficiency

(b) Which of the following is a common cause of electrical fires?
(i) Overloading circuits
(ii) Proper grounding
(iii) Regular maintenance
(iv) Low voltage

(c) What is the recommended type of electrical outlets for wet areas, such as bathrooms?
(i) Ungrounded outlets
(ii) GFCI outlets
(iii) Surge protected outlets
(iv) Three-pronged outlets

(d) What should you do if you experience an electrical shock?
(i) Pour water over the affected area
(ii) Apply a burn ointment immediately
(iii) Remove the person from the electrical source
(iv) Do nothing; it will subside on its own

(e) What does the term "overloading" mean in electrical safety?
(i) Using too many electrical devices simultaneously
(ii) Using energy-efficient appliances
(iii) Overcharging a battery
(iv) Insulating electrical wires

(f) What is the color coding for the ground wire in electrical systems?
(i) Red
(ii) Black
(iii) Green
(iv) Yellow

(g) What does the abbreviation "PPE" stand for in electrical safety?
(i) Personal Protective Equipment
(ii) Proper Power Efficiency
(iii) Protective Power Equipment
(iv) Professional Protection Evaluation

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(h) Which of the following should be done before working on electrical equipment?
(i) Wear insulated gloves
(ii) Use a metal ladder
(iii) Keep the equipment plugged in
(iv) Stand on a wet surface

(i) What is the purpose of a surge protector in electrical systems?
(i) To regulate the flow of electricity
(ii) To protect against voltage spikes and surges
(iii) To increase energy efficiency
(iv) To generate electricity

(j) What is the purpose of regular electrical equipment inspections?
(i) To increase energy efficiency
(ii) To ensure compliance with safety regulations
(iii) To generate electricity
(iv) To reduce the risk of electrical fires

*Q.2 State the necessity of earthing the electrical equipments and also mention the advantages of earthing the neutral point. **[14]**

*Q.3 Discuss Primary and secondary electrical shocks. Discuss possibilities of getting electrical shock and its severity **[14]**

Q.4 List the safety precaution during agricultural pump installation. Write some do's and don'ts for safety in the use of domestic electrical appliances. How the shock can be prevented from wet wall? **[14]**

*Q.5 Discuss the risks during installation of electric plant and equipment. List the safety precaution on installing large power transformer. **[14]**

*Q.6 What are the risks during installation of electric plant and equipment? State the steps in commissioning of electric plant. **[14]**

Q.7 What are the different hazardous zones? Name the different types of enclosures for hazardous locations. Mention few causes of sparks and flashover associated with electrical equipment. **[14]**

Q.8 List some electrical safety requirements in respect of: **[14]**
(a) Domestic/wiring/fittings
(b) Use of domestic electrical appliances.

Q.9 Write short notes on *any two* of the following: **[7x2=14]**
(a) Safety precautions to be taken during plant energizing.
(b) personal protective equipment for hazardous condition
(c) causes of sparks and flashovers in electrical plant and machines
(d) Fire Extinguishers

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